Alignment, Aid, and Aggression: Bringing Strategic Interdependence into the Two-Good Theory of Foreign Policy

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Why Foreign Policy Varies: The Puzzle

- Why are some countries more aggressive and interventionist than others?
- The *Two-Good Theory* (TGT) (Palmer and Morgan 2011) offers a compelling framework:
  - Countries optimize between **maintenance** (preserving liked aspects of the status quo) and **change** (revising disliked aspects)
  - Predictions: more powerful and more dissatisfied countries are more change-seeking
- **But TGT ignores strategic interdependence** — countries do not make foreign policies in isolation

The Gap in TGT Has Real Consequences

- Foreign policy is inherently a **general equilibrium** problem
- What one country does shapes what others can achieve
- Treating foreign policy as a partial equilibrium problem may yield:
  - Missed predictions
  - Incorrect ceteris paribus claims (e.g., Palmer and Morgan 2011)
- This study formalizes a **game-theoretic extension** to TGT that accounts for strategic interdependence and tests its implications

Countries Trade Off Change and Maintenance Under Scarcity

- Classic TGT utility: $U = Q_m^{\pi_1} \times Q_c^{\pi_2}$, subject to budget $b = Q_m + \Omega Q_c$
- Key parameters:
  - π_2/π_1: relative dissatisfaction (preference for change over maintenance)
  - b: power (resource endowment)
  - Ω: comparative cost of change (decreases with power)
- Optimal demands: $Q_m = \pi_1 b$, $Q_c = \pi_2 b/\Omega$

Foreign Policy Alignment Links Countries' Choices

- Extended model: country i's realized change and maintenance depends on j's activities:

$$Q_{ic} = q_{ic} + \alpha q_{jc}$$
$$Q_{im} = q_{im} + q_{jm}$$

- $\alpha \in (-1, 1)$: **foreign policy alignment**
  - $\alpha \to 1$: aligned — others' change-seeking helps i
  - $\alpha \to -1$: adversarial — others' change-seeking harms i
- Maintenance acts as a **public good**; change is a **rival good** *or* **public good** shaped by alignment
- A unique Nash Equilibrium exists (by Rosen's (1965) concavity proof)

Monte Carlo Analysis Yields Three Predictions

- Monte Carlo analysis: 10,000 simulated international systems, $N = 10$ countries each
- Exogenous parameters drawn randomly: alignment α, power b, dissatisfaction π_2

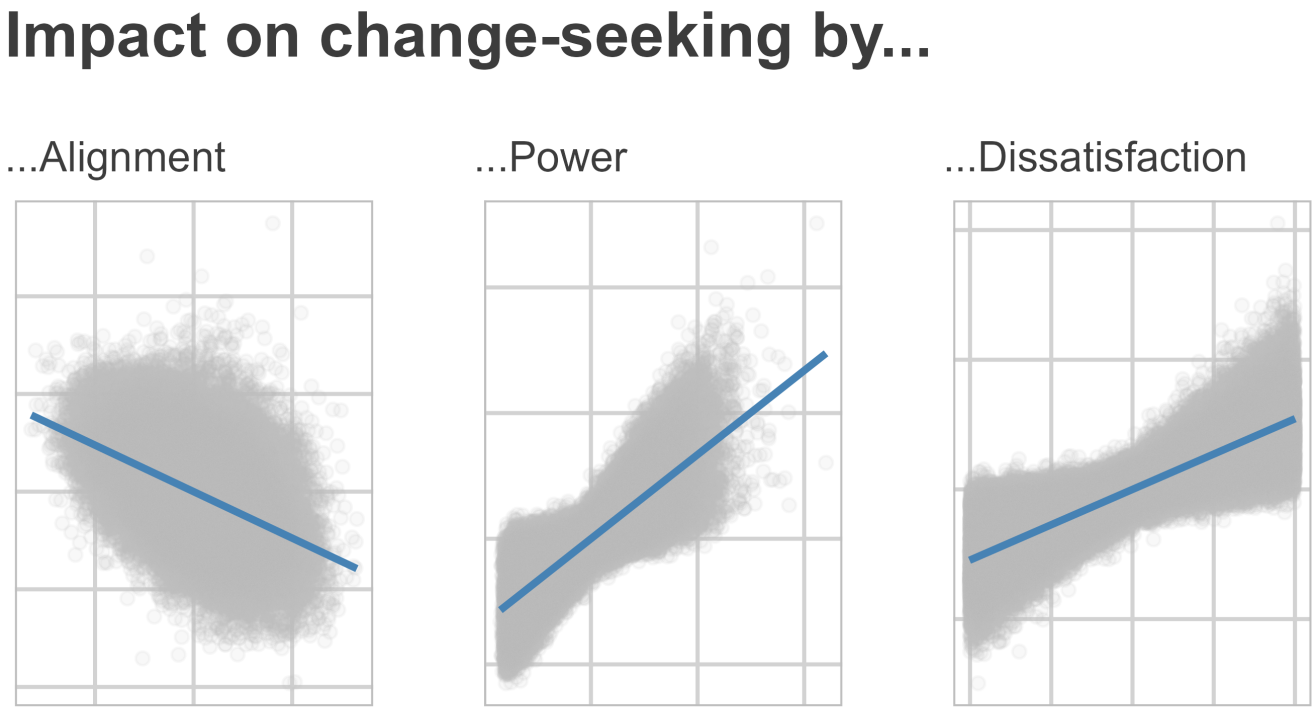

Figure 1: Partial regression slopes from Monte Carlo analysis

Monte Carlo Analysis Yields Three Predictions (cont.)

- Predictions from endogenized TGT:
  - H_1: Power $\uparrow \rightarrow$ change-seeking $\uparrow$
  - H_2: Dissatisfaction $\uparrow \rightarrow$ change-seeking $\uparrow$
  - H_3: Alignment $\uparrow \rightarrow$ change-seeking $\downarrow$
- Intuition: aligned countries can free-ride on each other's efforts to change the status quo; misaligned countries must work harder to move the system in their preferred direction
- This *reverses* the informal suggestion by Palmer and Morgan (2011) that higher threat leads to more maintenance-seeking

Two Change-Seeking Behaviors Are Tested

- **Outcome 1**: MIC (Militarized Interstate Confrontation) initiation (1946–2014) (Gibler and Miller 2024)
  - Binary: did a country initiate a conflict involving use of force?
- **Outcome 2**: Total bilateral ODA commitments (1966–2023, OECD DAC3a)
  - Continuous: millions of 2023 constant USD
- Both are argued by Palmer and Morgan (2011) to reflect change-seeking foreign policy

Key Explanatory Variables

Concept	Measure	Source
Power	Surplus Domestic Product (SDP), share of global total	Anders, Fariss, and Markowitz (2020)
Dissatisfaction	Log ratio of expected to realized benefits	Kent (2020)
Alignment	Average UN GA voting similarity (liberal order orientation)	Bailey, Strezhnev, and Voeten (2017)

- Models: **Logit** (MIC initiation) and **PPML** (ODA commitments)
- Both include random country slopes and cubic time/peace spell trends

Results Consistent with Theory

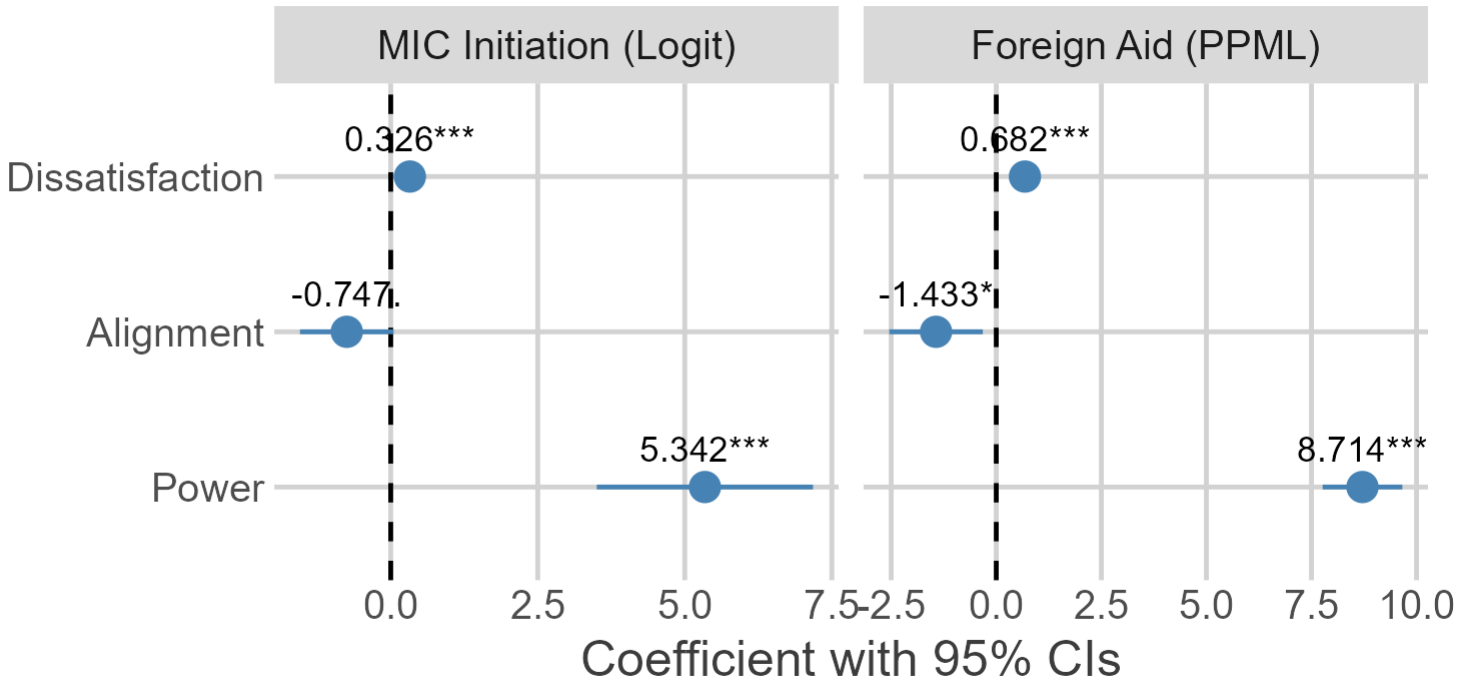

Figure 2: Model Estimates

- Power and dissatisfaction: positive, significant in both models
- Alignment: negative in both models; significant for ODA, *marginally* significant for MIC

Effects Are Substantively Large

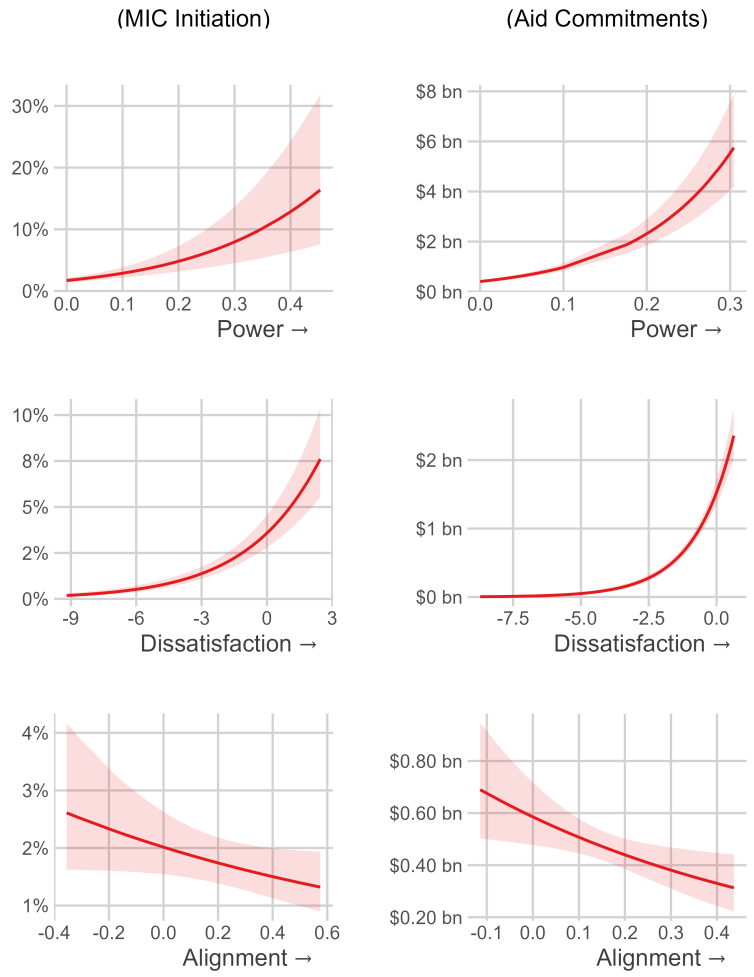

Figure 3: Predicted marginal effects

- Power: going from near-zero to 40%+ of global SDP → MIC probability rises from ~0% to ~15%
- Dissatisfaction: low to high → MIC probability rises from ~0% to ~8%; ODA rises from ~$0 to ~$2B
- Alignment: low to high → MIC probability falls from ~2.5% to ~1.3%; ODA falls from ~$0.7B to ~$0.3B

TGT Explains Conflict and Aid Better When Adjusting for Alignment

- Results support all three hypotheses derived from the endogenized TGT
- Consistent with classic TGT (power and dissatisfaction $\rightarrow$ change-seeking)
- New prediction confirmed: **alignment reduces change-seeking**
- Alternative theories (Gilpin 1981; Braumoeller 2013) cannot fully account for the independent effects of power and alignment

Future Work Should Extend These Findings

- Limitations and open questions:
  - Better measures of systemic dissatisfaction needed
  - Maintenance-seeking outcomes (MIC reciprocation, defense spending) remain difficult to test cleanly due to selection bias and conceptual overlap
  - No dynamic theory of how foreign policies reshape the status quo over time
- Broader implication: as U.S. foreign aid collapses and bellicosity rises, **substitution between aid and conflict** deserves urgent attention
- General equilibrium approaches to IR are underutilized — this study shows their value

References I

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